

Alex Sareh

+1 (239) 671-6638 | asareh3@gatech.edu

Introduction

Hello! I'm an undergraduate student at Georgia Tech, pursuing a Bachelor of Science in Computer Engineering with concentrations in Computing Hardware & Emerging Architectures and Systems & Architecture. I am highly adaptable, a strong communicator, and can work great in both solo and collaborative environments. I am seeking a Summer 2027 internship.

Education

Georgia Institute of Technology | Atlanta, GA

Fall 2024 – Present

Bachelor of Science in Computer Engineering, GPA 3.84

Expected Graduation, Fall 2027

Concentrations in **Computing Hardware & Emerging Architectures** and **Systems & Architecture**

Skills

Software: C, C++, Java, Python, RISC-V Assembly, Embedded Systems (Arduino, ESP32)

Hardware: Verilog, VHDL, PCB design (KiCad), schematic capture, soldering

Platforms: Git, Linux, Windows

Experience

The Hive – Peer Instructor

Sep 2025 – present

- Volunteer time to help students bring projects to life using the electronics benchtop, 3D printers, laser cutters, and other equipment.
- Support student learning by answering technical questions and demonstrating best practices.

Programming HW/SW Systems – Teaching Assistant

Jan 2026 – present

- Class covers Low level C/C++, ISA datapath, Risc-V assembly, and embedded systems programming.
- Support a large undergraduate course by holding weekly open office hours to support students with programming and hardware/software system concepts.
- Answer student questions related to assignments, labs, and system-level design.
- Provide constructive feedback through assignment grading to improve student understanding and performance.

retroTECH – Student Assistant

Sep 2025 – present

- Staff open lab hours for Georgia Tech's computing history and hardware preservation lab.
- Monitor and maintain legacy computing equipment during student use.
- Assist users with safe operation, setup, and basic troubleshooting of hardware systems.

RoboJackets – Electrical Engineering Trainer

Aug 2025 – Dec 2025

- Prepped new RoboJacket members for working on electrical hardware.
- Curriculum starts with basic electronics concepts and prototyping and then transitions to printed circuit board (PCB) design and overall design fundamentals.

IMAG History & Science Center – Camp Counselor Volunteer

Summer 2023

- Led and supervised K-8 students and organized daily STEM-focused activities and experiments.
- Communicated technical ideas in an age-appropriate, engaging manner.

Projects

Control Voltage Touch Controller: Designed and built a custom PCB-based control voltage touchplate for an analog synthesizer. Did the circuit design, PCB fabrication, and soldering.

500g Sumo Robot: Built a 500-gram autonomous sumo robot with a team of 3 other people. I was the sole electrical engineering member, where I designed a custom PCB, soldered the components, and troubleshooted problems.

RISC-V Single-Cycle Processor: Designed and implemented a single-cycle RISC-V RV32I processor in Verilog, including instruction decode, ALU, register file, and memory access, verified through simulation tests.

Relevant Coursework

Architecture, Systems, Concurrency and Energy in Computation: The organizational principles of the major components of a processor, e.g., the core, memory hierarchy, I/O subsystem and basic operating system constructs.

Design & Analysis of Algorithms: Techniques of design and analysis of efficient algorithms for standard computational problems. Specifically discusses Divide and Conquer, Graph Algorithms, Dynamic Programming, and NP-completeness.

Programming HW/SW Systems: Creation of complex execution and storage mechanisms, based on instruction set architecture, for software design including high-level programming languages and operating systems.

Digital Design Laboratory: Design and implementation of digital systems, including a team design project. CAD tools, project design methodologies, VHDL, logic synthesis, and assembly language programming.